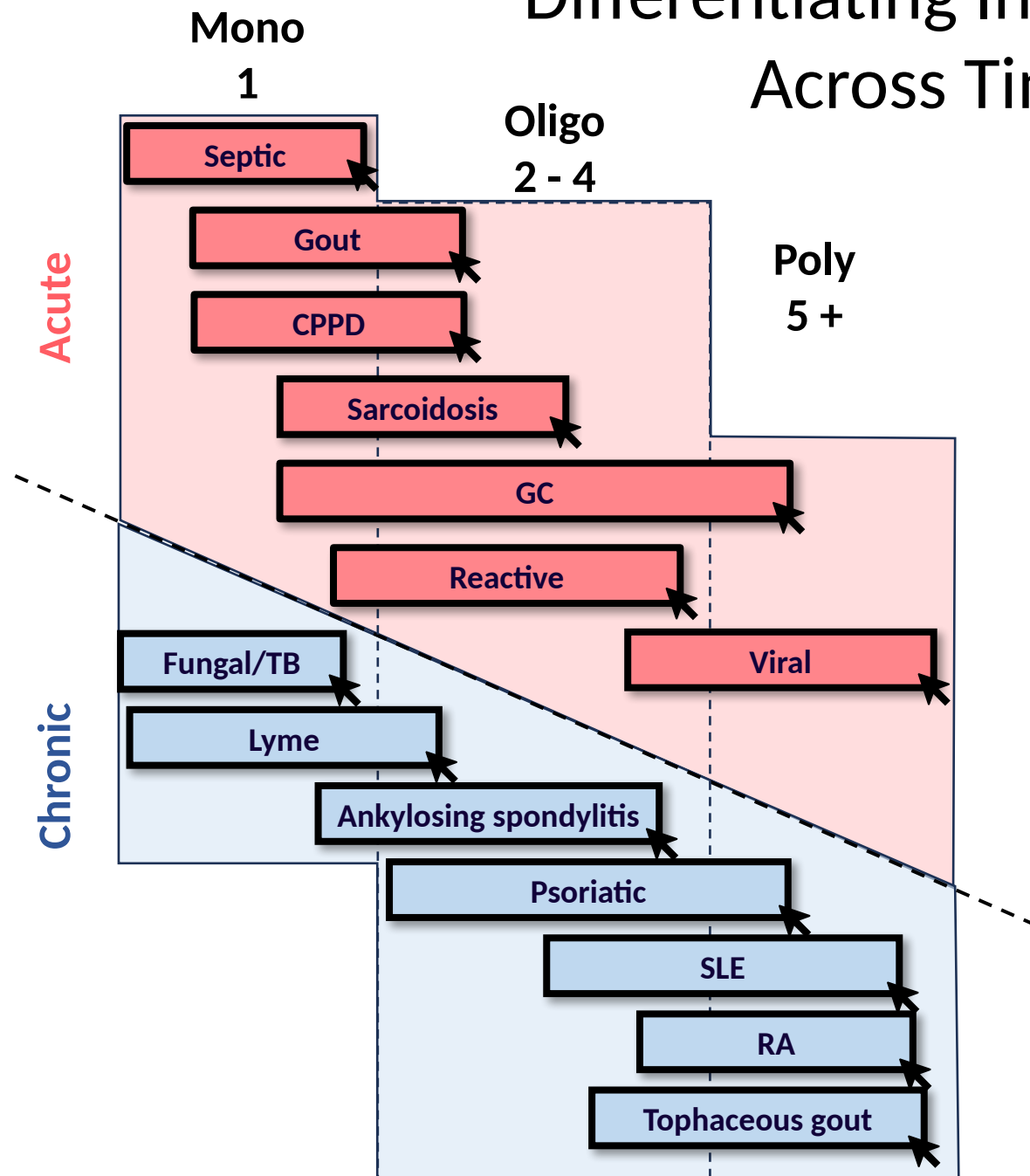
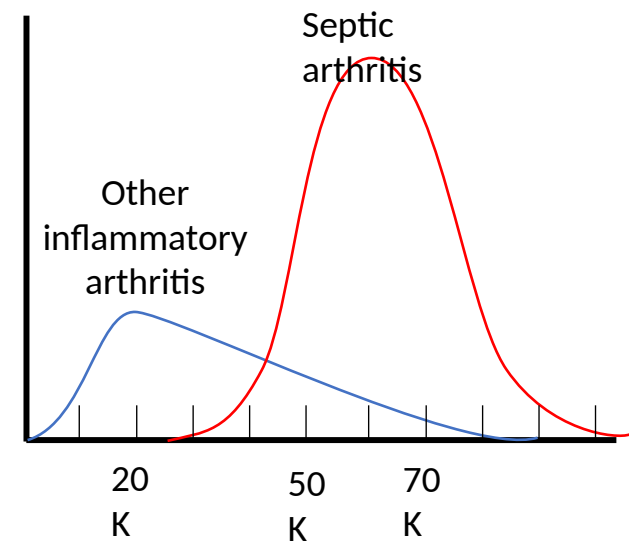


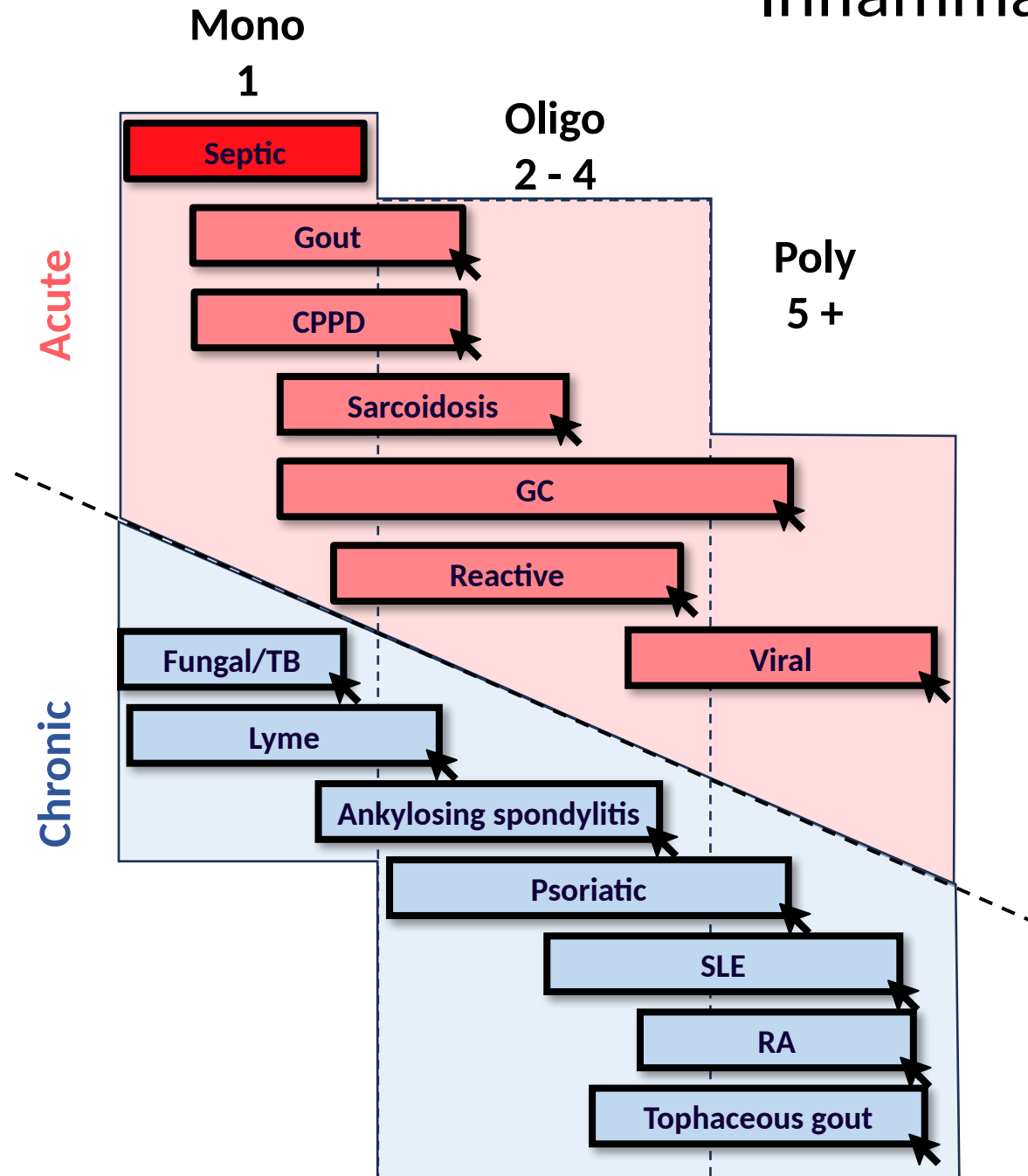
Differentiating Inflammatory Arthritis Across Time and Space



Synovial Fluid WBC Distribution



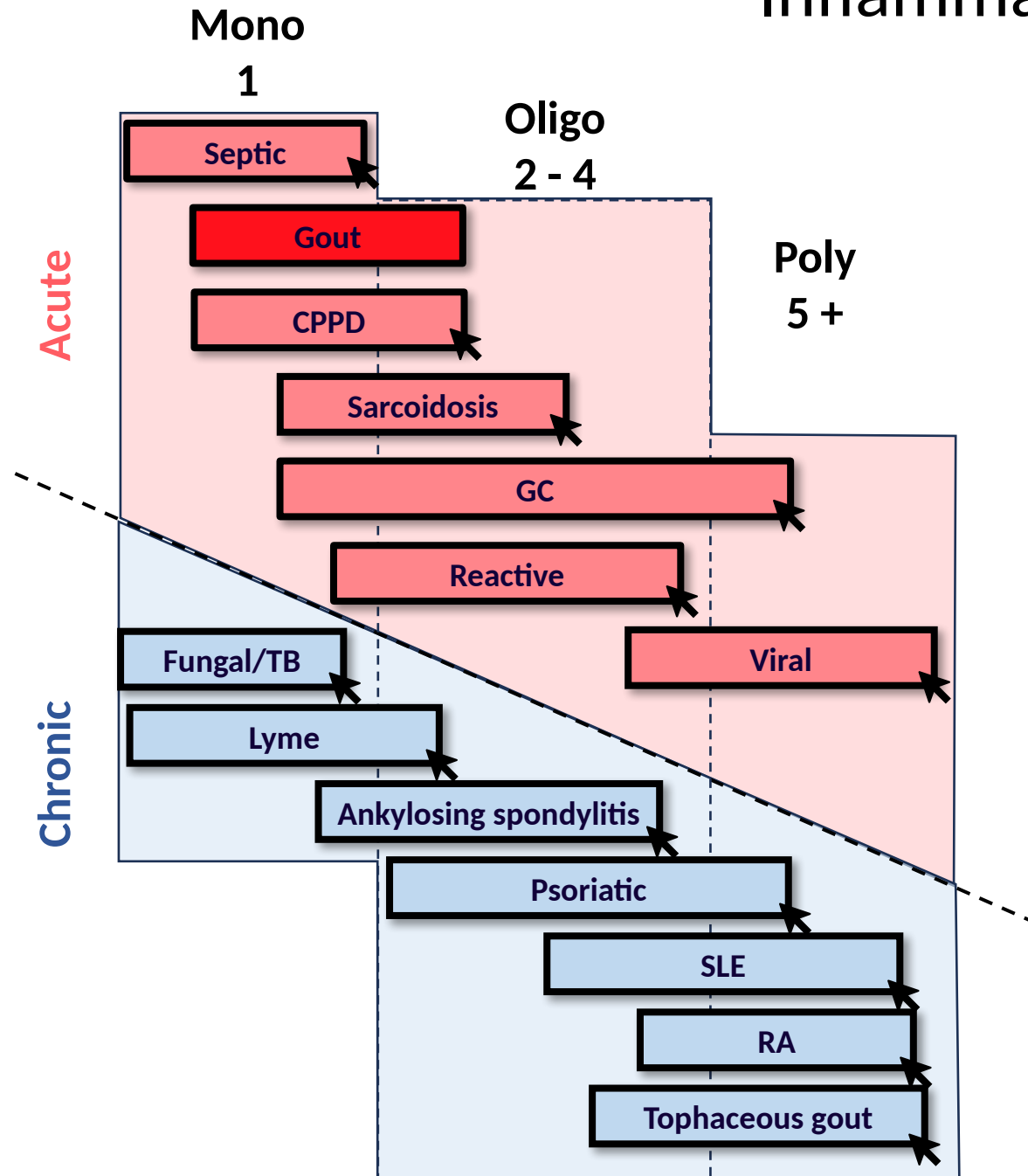
Inflammatory Arthritis



Septic arthritis

- Synovial fluid: typically >50K, PMN predominant (lower cutoff in prosthetic joints or after antibiotics)
- Empiric therapy based on gram stain: Vancomycin for GPC and ceftriaxone for gram-negative
- Urgent surgical washout is often necessary

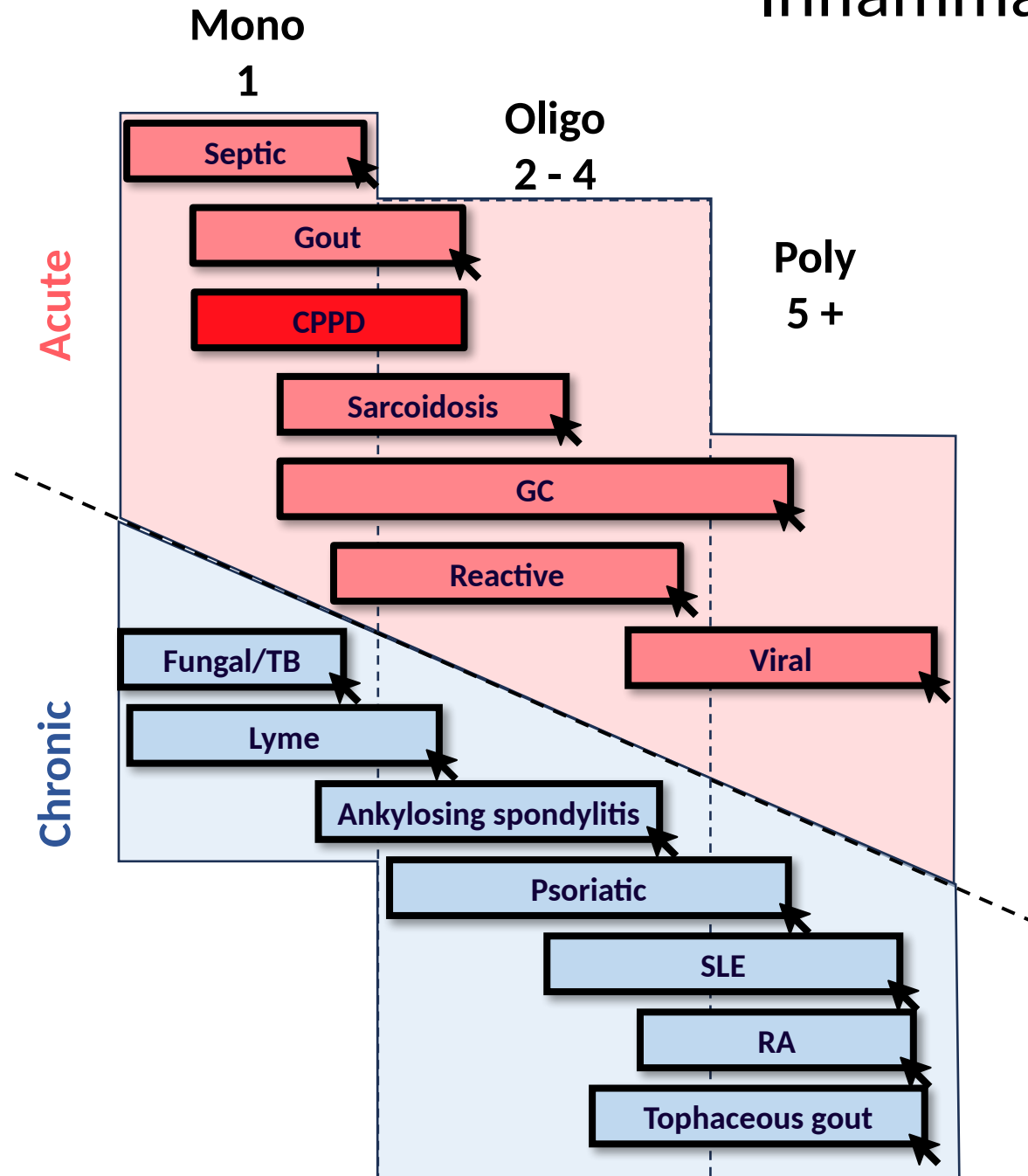
Inflammatory Arthritis



Acute Gout

- Synovial Fluid: monosodium urate crystals under polarized microscope
- MTP is the most common. Lower extremity predominant (including mid-foot). Can be any joint.
- Risk Factors: obesity, hypertension, and chronic kidney disease
- NSAIDs, colchicine, or corticosteroids for acute flares; urate-lowering therapy for chronic management
- Must take prophylaxis while escalating urate lowering therapy

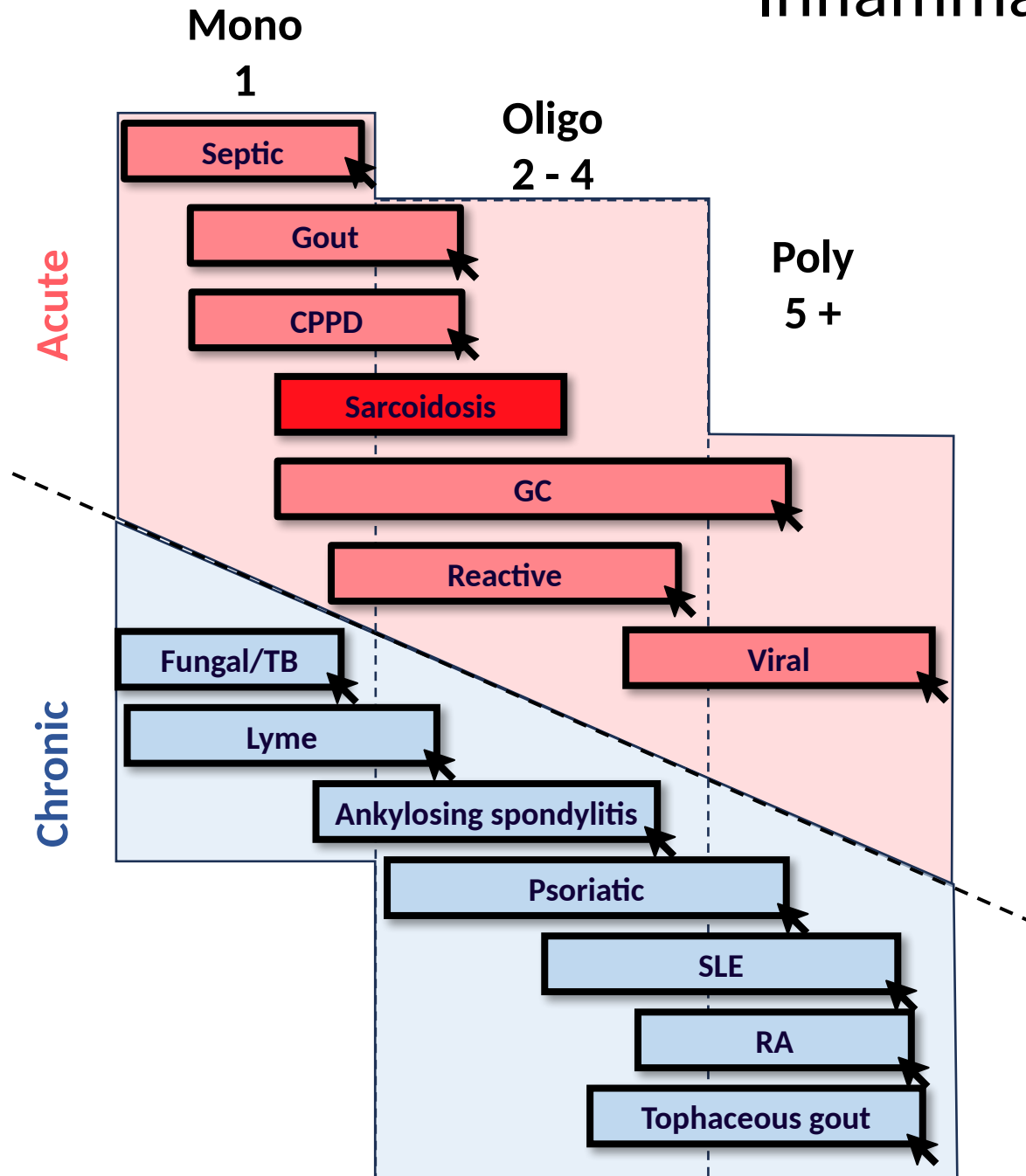
Inflammatory Arthritis



Calcium Pyrophosphate Dihydrate Deposition Disease (CPPD)

- Synovial fluid: CPP crystals
- Imaging: may show chondrocalcinosis (knees, wrists and fingers)
- Phenotypic Variability: Can present as recurrent acute arthritis, chronic inflammatory arthritis, or asymptomatic chondrocalcinosis
- Risk Factors: hyperparathyroidism and hemochromatosis
- NSAIDs, colchicine, and corticosteroids for acute attacks.

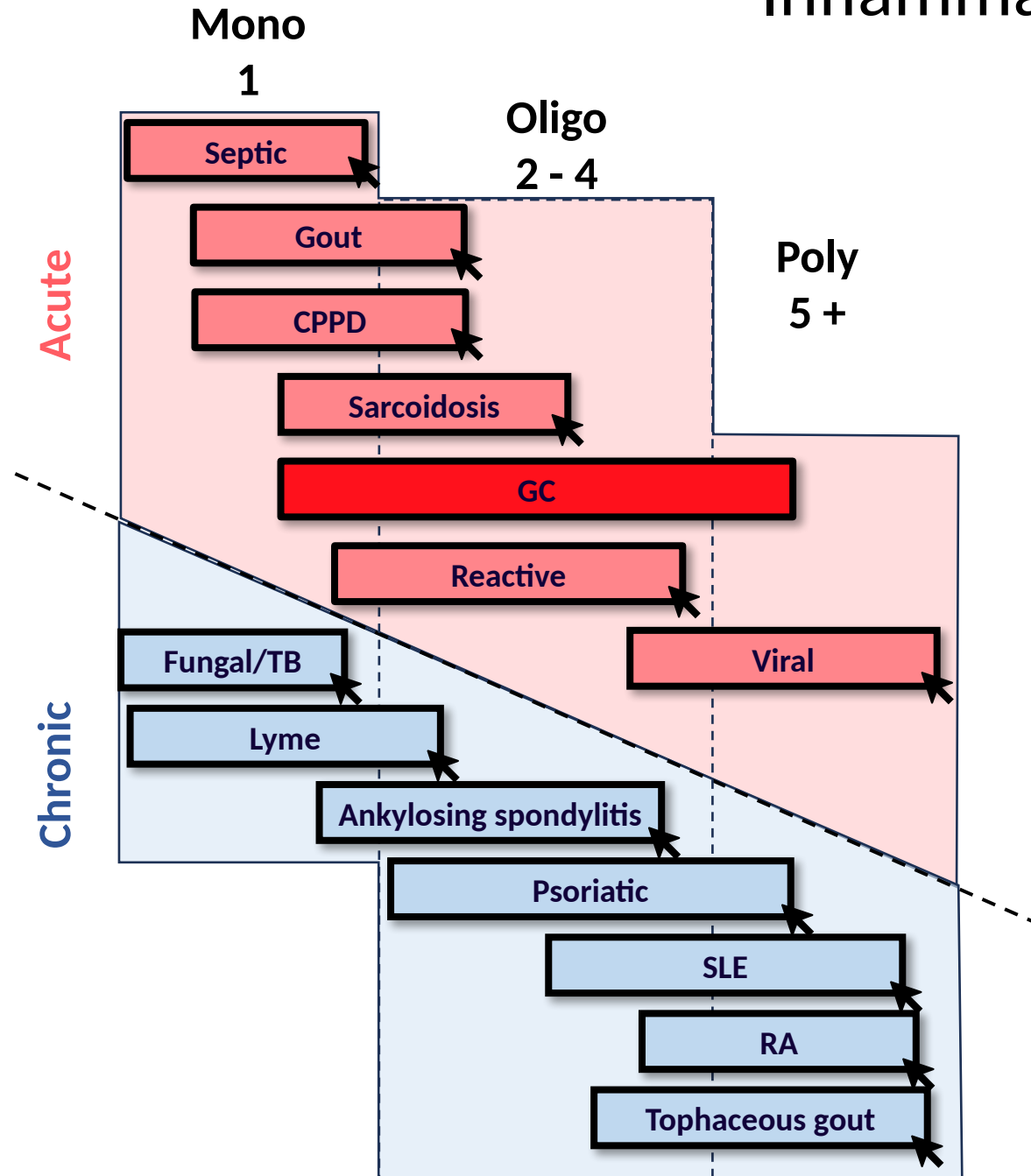
Inflammatory Arthritis



Sarcoidosis

- Co-exists with thoracic disease
- Acute: typically ankle, knee and wrist
- Lofgren's syndrome: Erythema nodosum + bilateral hilar adenopathy + acute arthritis
- Chronic sarcoid arthropathy is less common
- Treatment: NSAIDs, other sarcoid DMARDs (e.g., methotrexate, TNFi)

Inflammatory Arthritis



Gonococcal Arthritis

- Triad: migratory polyarthralgia, tenosynovitis, and pustular dermatitis
- Synovial fluid: typically 10-50K WBCs.
- *Neisseria gonorrhoeae* can be isolated from synovial fluid, blood, or mucosal sites.
- Ceftriaxone is 1st line
- Add doxycycline or azithromycin for potential co-infection with *Chlamydia trachomatis*
- Good prognosis with appropriate antibiotics

Gonococcaemia



Gonococchemia



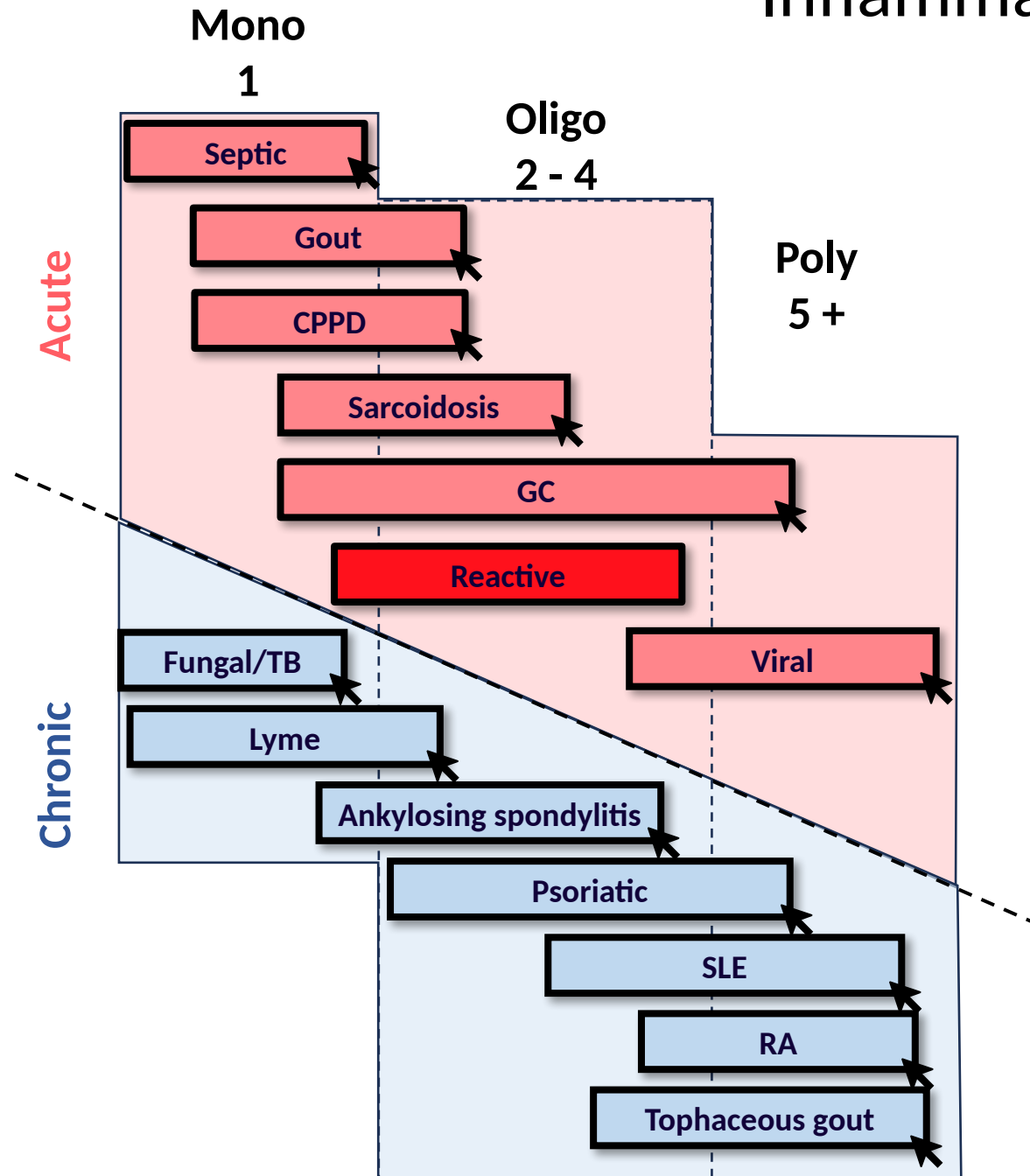
dermnetnz.org



Newman CR, Am J Med.
2021

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Inflammatory Arthritis



Reactive Arthritis

- Typically follows a GI or GU infection by 1-6 weeks
- Asymmetric oligoarthritis, often involving the lower extremities
- Strong genetic predisposition, particularly carriers of HLA-B27 allele
- Extra-articular manifestations: uveitis, urethritis, and skin lesions (e.g., keratoderma blennorrhagicum)
- Negative synovial fluid cultures

Keratoderma



Keratoderma blennorrhagicum – associated with reactive arthritis



[Reactive arthropathy \(dermnetnz.org\)](http://dermnetnz.org)



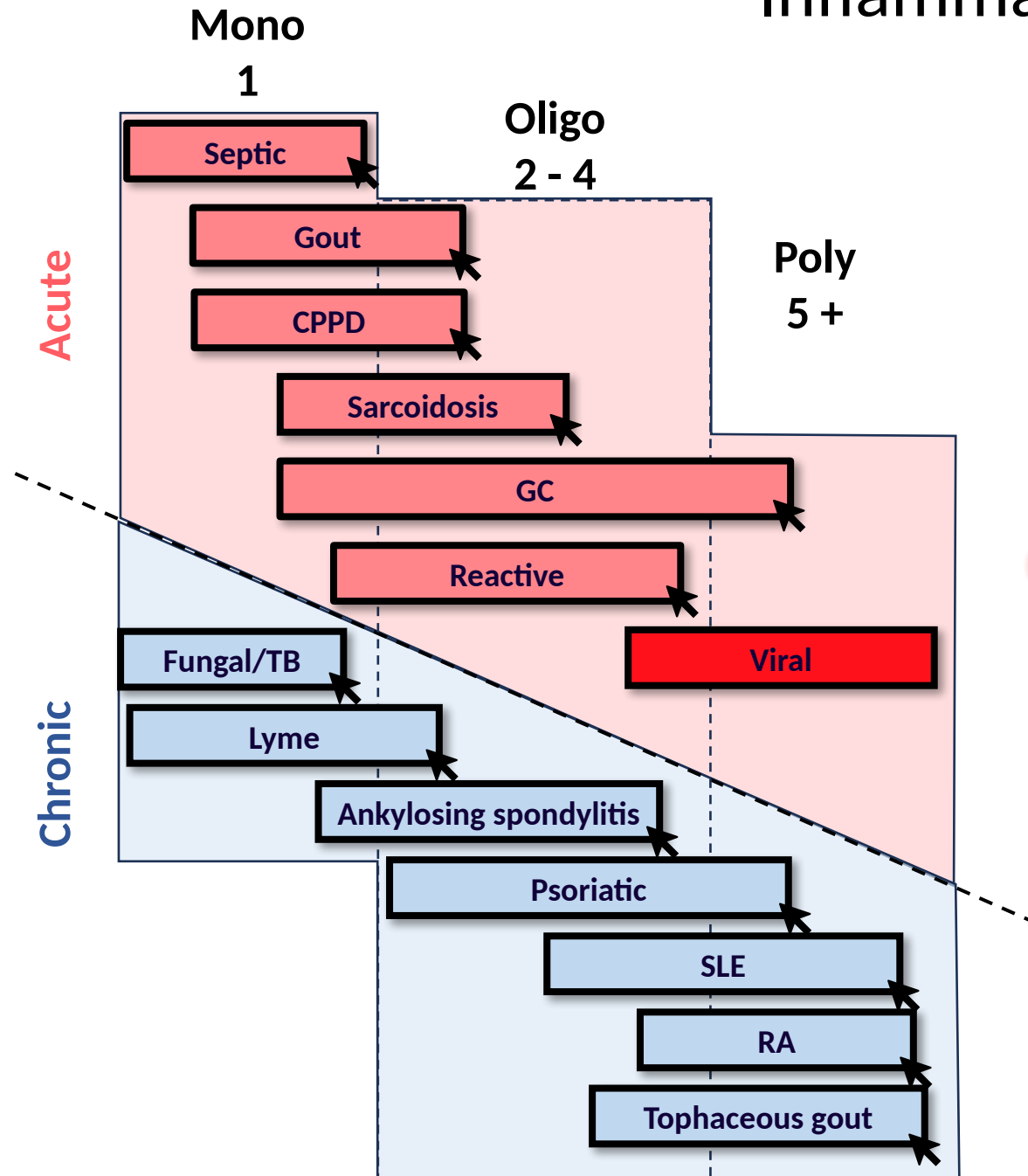
Coelho I. BMJ Case Rep. 2017



Florell SR, N Engl J Med. 2003



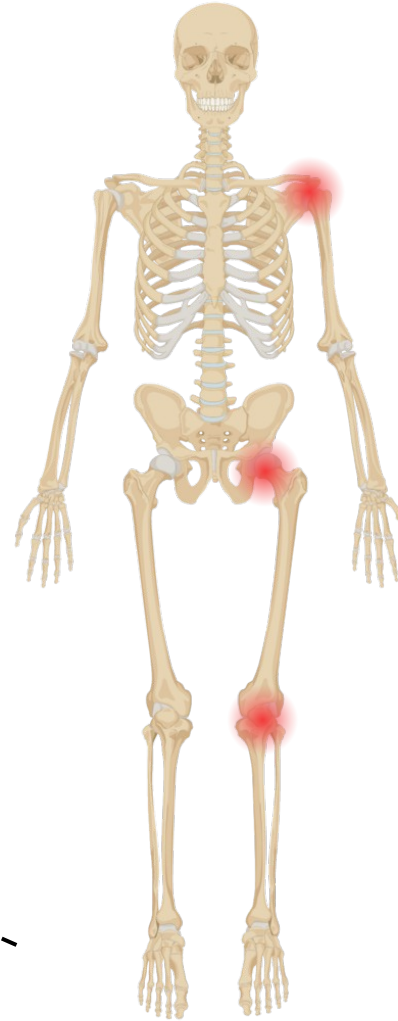
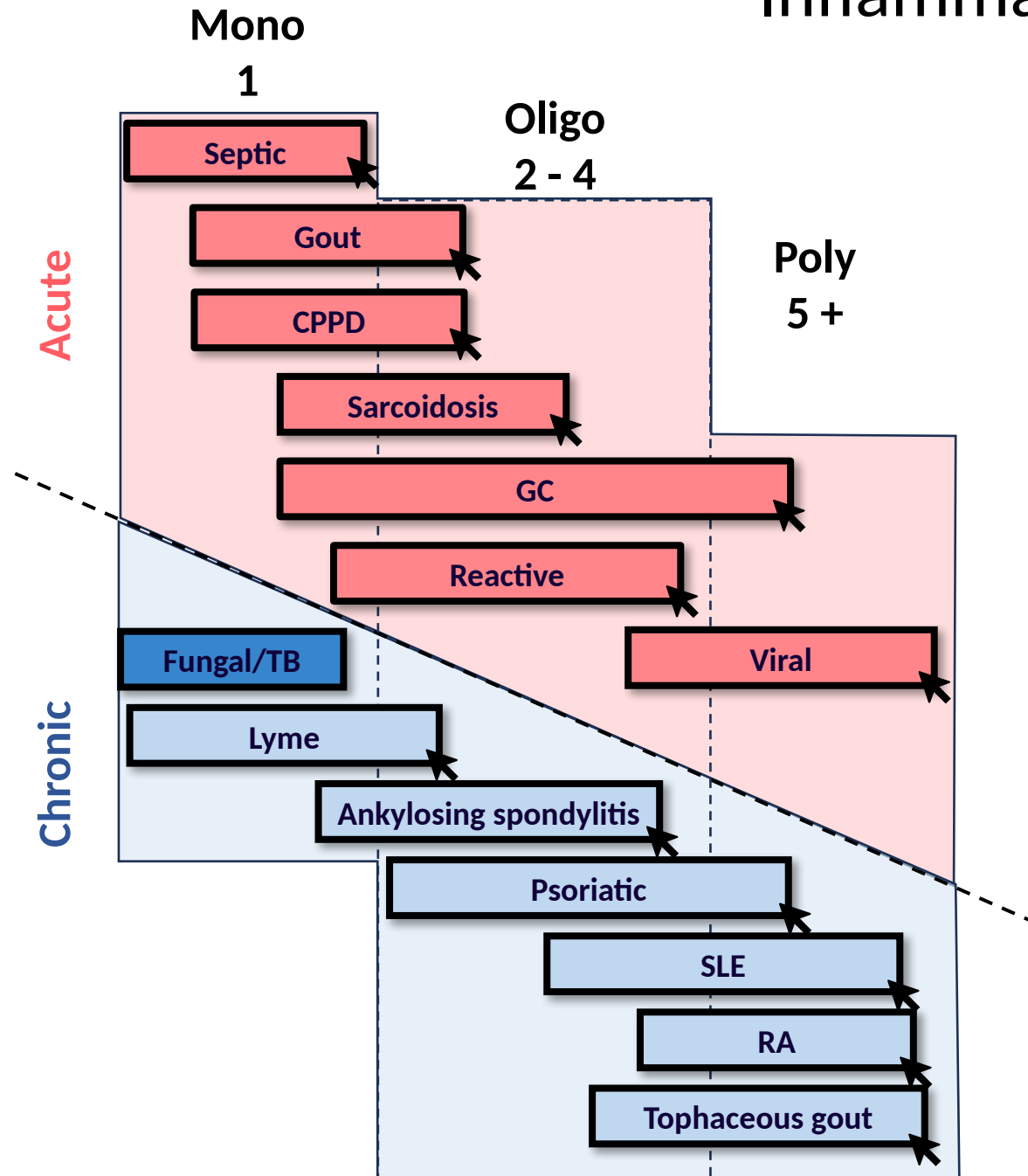
Inflammatory Arthritis



Viral arthritis

- Variable presentation based on causative virus
- Parvovirus B19 (most common, RA-like symmetric small joint) → high IgM antibodies (IgG = prior exposure)
- Hepatitis B or C (next most common).
- Travel related include Dengue, Chikungunya, Zika.
- Some lead to chronic arthritis (>6 weeks)
- Manage with NSAIDs. Consult with ID for other possible manifestations of travel associated viruses
- Many may induce transient autoantibody formation (ANA, RF, etc) so less helpful to check for acute arthritis

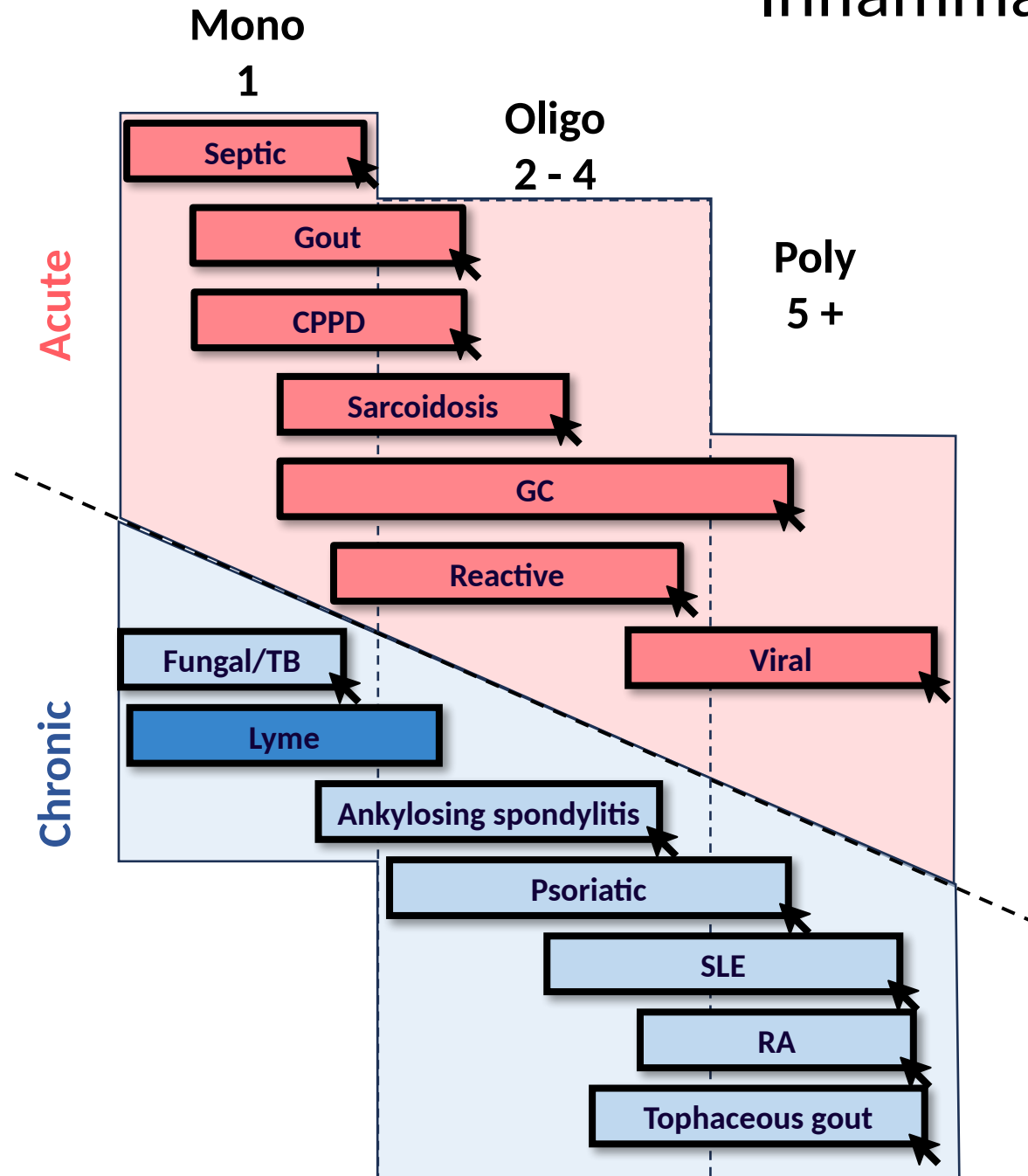
Inflammatory Arthritis



Fungal and Tuberculosis

- Indolent
- Risk factors: immunosuppression, travel to endemic areas
- Synovial fluid: typically 10-20K WBC, may be culture negative (consider synovial biopsy)
- Imaging: may show erosions
- Treatment: Surgical evaluation for wash out and infectious disease consult

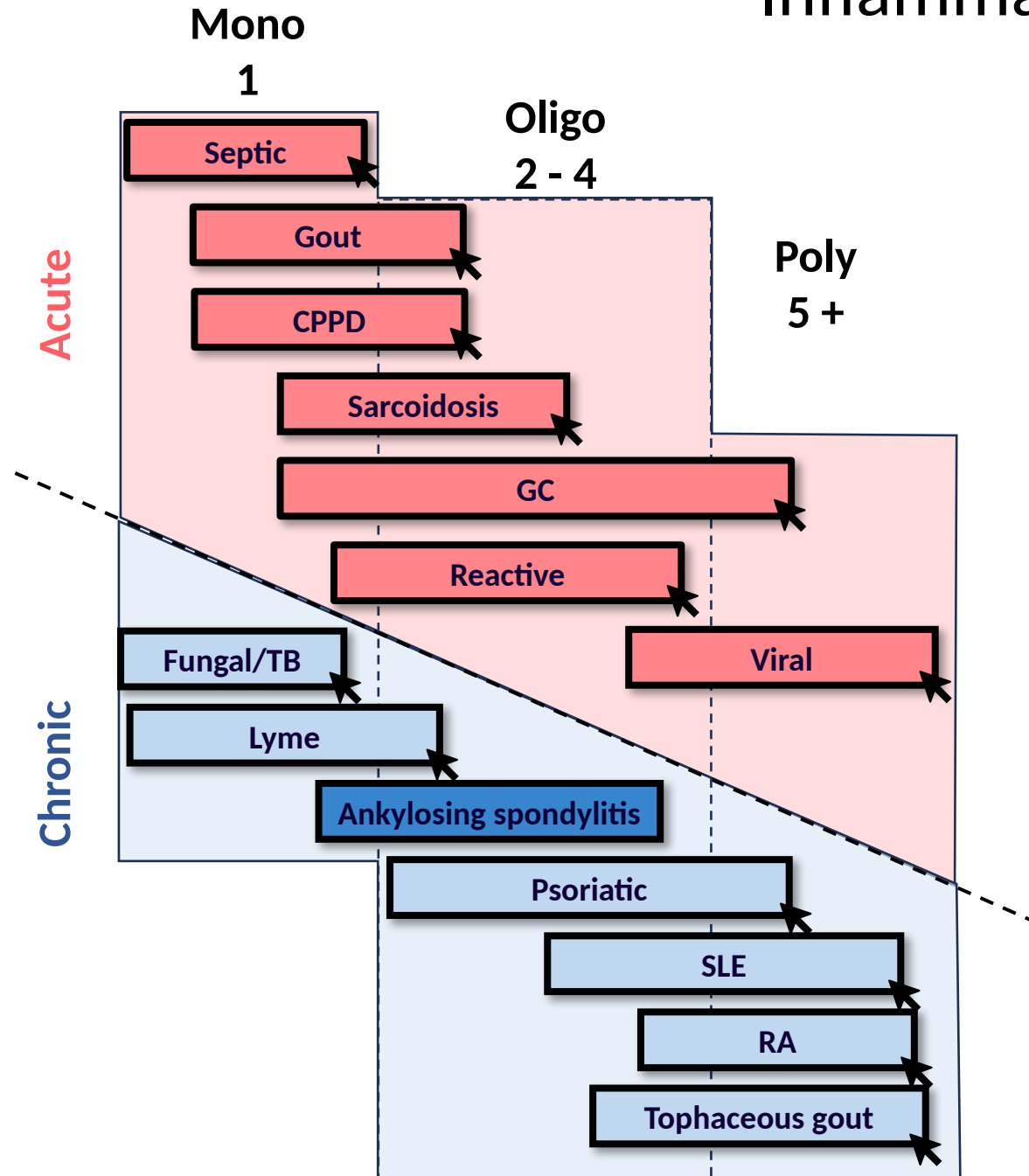
Inflammatory Arthritis



Lyme disease

- Caused by spirochete *Borrelia burgdorferi*, typically months after initial tick bite
- Risk factors: Outdoor exposure in endemic area
- Often affects a large joint (knee) but can affect elbow, shoulder, ankles, wrists
- Synovial fluid: >2K WBC, send for *Borrelia* PCR
- Treatment: typically, 4-week course of doxycycline
- May lead to chronic arthritis requiring DMARDS

Inflammatory Arthritis



Ankylosing spondylitis

- Other spondyloarthropathies (SpondA) include, reactive arthritis (usually acute, see other tab), psoriatic arthritis (see other tab), IBD-associated arthritis and share a genetic association with HLA B27
- SI involvement > other spine. May have peripheral involvement less symmetric than RA or lupus
- Extensive ROS to differentiate SpondA: rashes, GI symptoms, axial symptoms, eye symptoms, tendon insertion pain/swelling, dactylitis, recent infectious symptoms, dysuria
- ESR, CRP, HLA B27; consider RF and CCP
- Imaging (include SI joint): soft tissue swelling, marginal erosions, syndesmophytes, ankylosis
- Oral DMARDs (e.g., sulfasalazine, MTX) and TNFi for axial involvement.

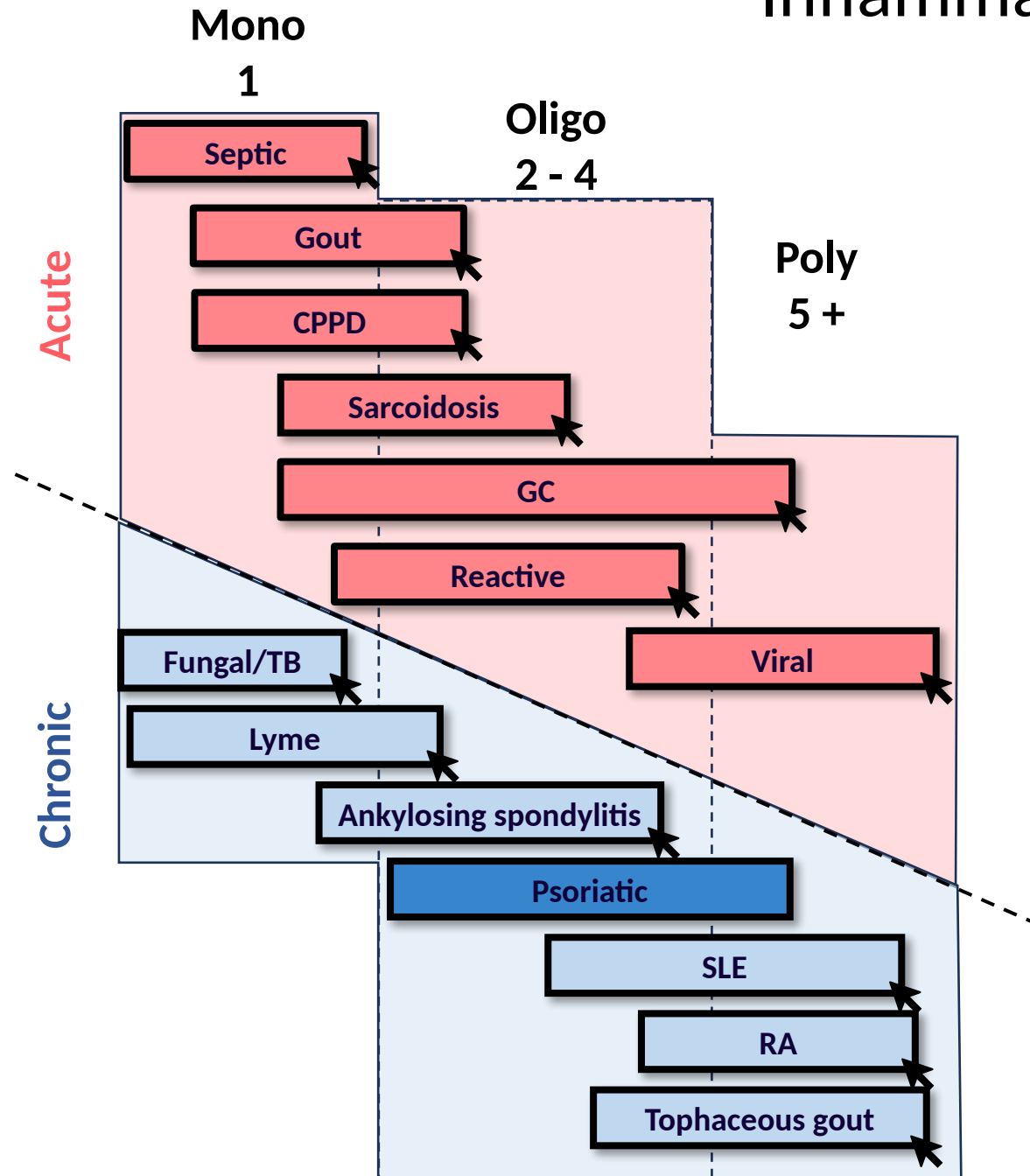


Ankylosing spondylitis



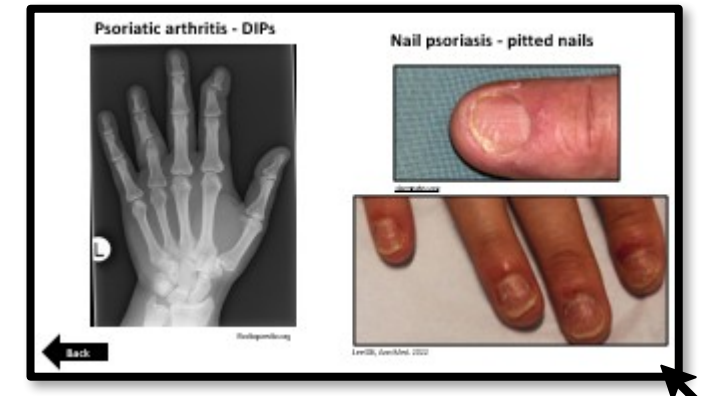
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Inflammatory Arthritis



Psoriatic

- Polyarticular > Oligoarticular
- Asymmetric, axial (spine), peripheral joints (including DIP)
- Arthritis severity does not correlate with derm severity
- Periarticular disease: enthesitis (tendon insertion pain), dactylitis
- Extraarticular disease: Nail changes (onycholysis, pits), skin exam, screen for inflammatory eye disease and IBD symptoms.
- Labs: consider RF/CCP, uric acid, joint aspiration if able
- Imaging: hand/foot and SI joint (if axial symptoms)



Psoriatic arthritis - DIPs



Radiopaedia.org

Nail psoriasis - pitted nails



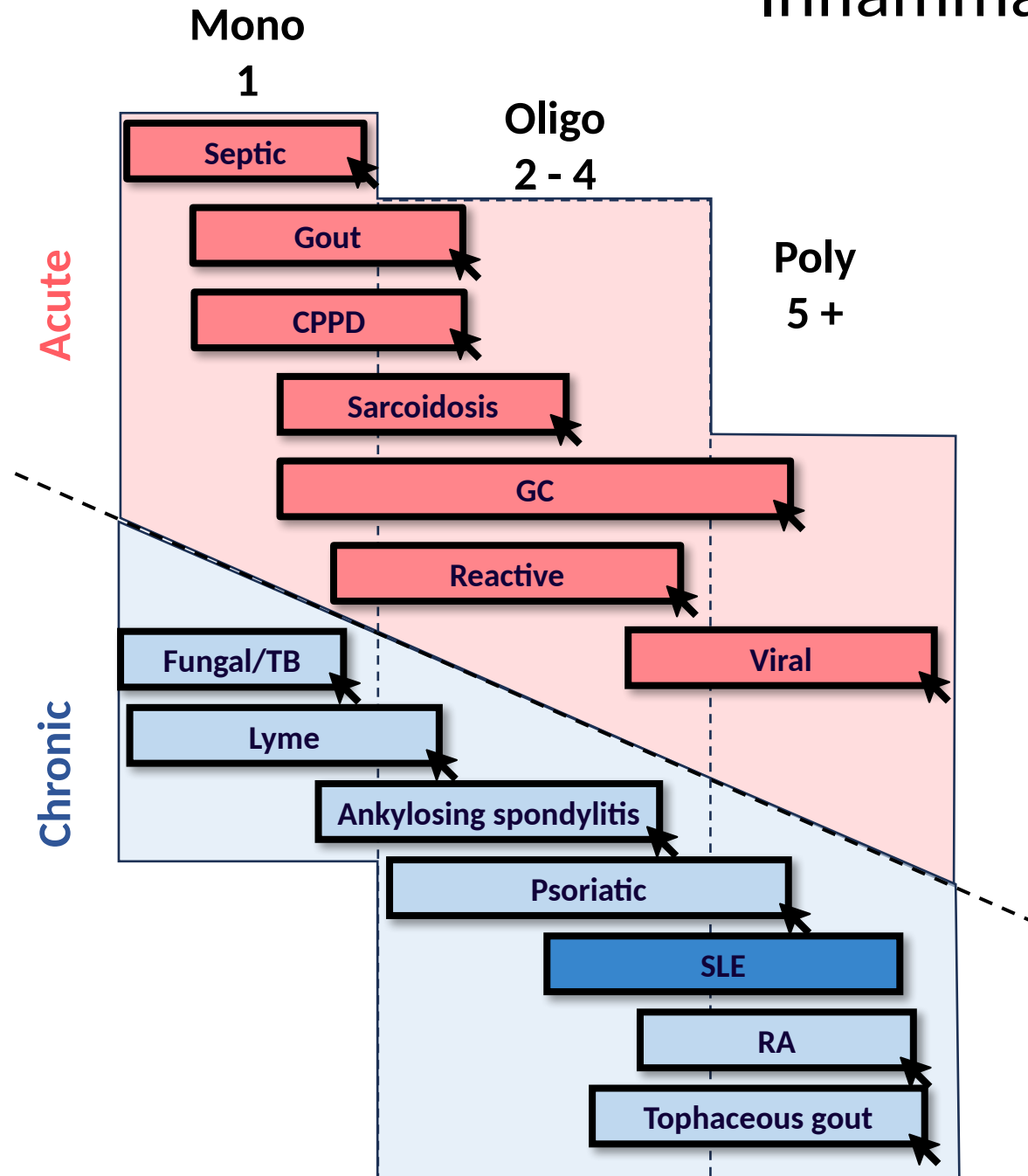
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Lee DK, Ann Med.
2022

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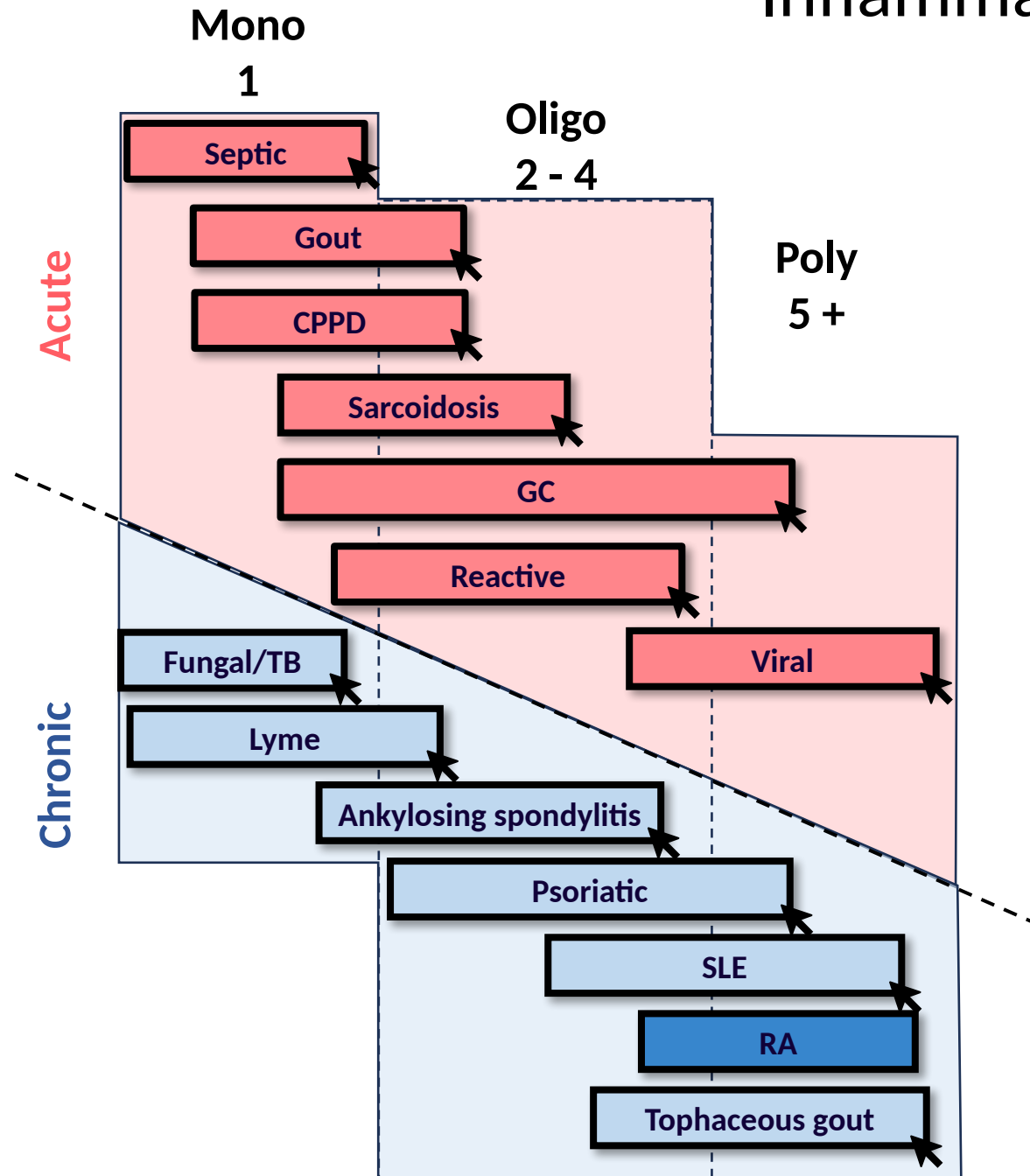
Inflammatory Arthritis



SLE

- Small joint predominant, symmetric
- Other SLE ROS: sun sensitive rashes, oral ulcers, sicca symptoms, alopecia, chest pain/DOE, Raynaud's
- NSAID or low dose prednisone responsive. Long term management: HCQ, methotrexate, and azathioprine are commonly used for joint pain in lupus
- ANA+. Monitor disease activity with CBC w/diff, C3, C4, dsDNA, ESR, CR, UA, urine protein/cr.
- Imaging: may show soft tissue swelling, periarticular osteopenia. Erosions are less common than RA and psoriatic.

Inflammatory Arthritis



Rheumatoid arthritis

- Small joint predominant, symmetric
- Typical involvement of wrists, MCPs, PIPs, ankles, MTPs. Spares the DIPs, thoracic and lumbar spine
- RF, CCP, ESR, and CRP
- Imaging: soft tissue swelling, periarticular osteopenia, marginal erosions. X-rays of hands and feet at diagnosis and every 1-2 years to assess for erosions
- Treatment: acute management of pain with NSAIDs and prednisone. First line DMARD is MTX.



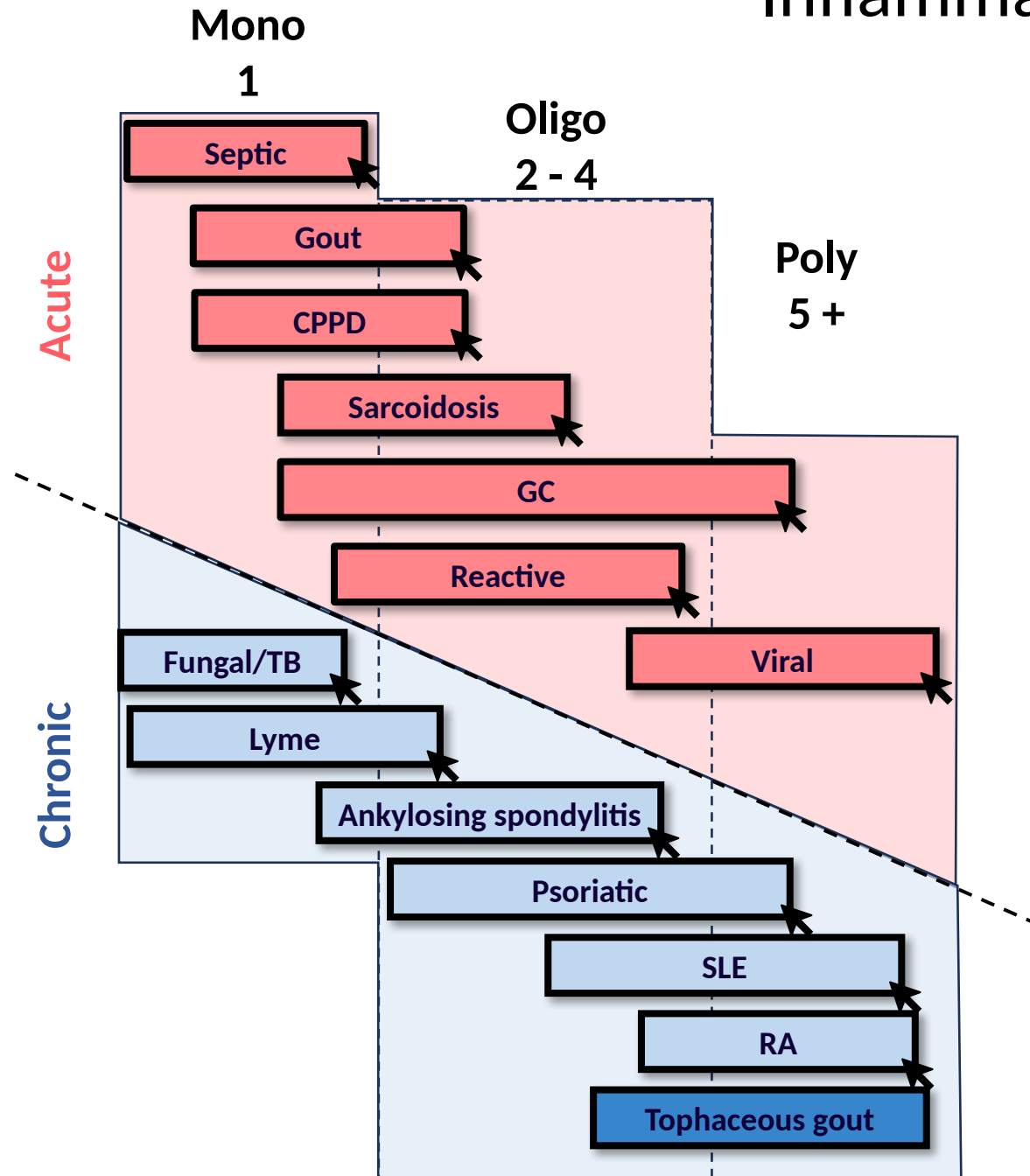
Rheumatoid arthritis – wrists, MCPs, PIPs



Case courtesy of Suman Bandhu, Radiopaedia.org,
rID: 82524

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Inflammatory Arthritis



Tophaceous gout

- Continual nidus of inflammation -> tophus (soft tissue nodules)
- Commonly with chronic arthritis punctuated by acute monoarthritis episodes
- Favors cool joints (fingertips, toes, Achilles, elbows, space distal to patella)
- Serum uric acid >7 (may be normal during flares)
- Imaging: radiolucent tophi, punched out erosions with overhanging edges especially in hands and feet
- Treatment: uric acid lowering therapy (allopurinol is 1st line) plus prophylaxis (e.g., low dose colchicine) while titrating along with flare action plan. Goal UA <6.

Tophaceous gout

Favors cool joints - fingertips, toes, Achilles, elbows, space distal to patella



Case courtesy of Imran Ahmad Khan, Radiopaedia.org, rID: 85874

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Take-Home Points

1. Start with time and space — acuity (acute vs chronic) and joint count (mono-/oligo-/polyarticular) rapidly narrow the differential for inflammatory arthritis.
2. Never miss infection — acute monoarthritis is septic until proven otherwise. Synovial fluid WBC + crystals + cultures, timely antibiotics, and surgical consultation are critical.
3. Pattern recognition matters — symmetry, axial involvement, DIP disease, enthesitis, dactylitis, and extra-articular features (skin, eye, GI, GU) differentiate RA, SLE, psoriatic arthritis, and spondyloarthropathies.
4. Match treatment to etiology and trajectory — acute flares need anti-inflammatory therapy; chronic autoimmune and indolent infectious arthritis need early disease-modifying treatment and specialty involvement.
5. Gout ULT: allopurinol first-line; start after >2 flares/yr (or after 1 flare if CKD ≥ 3 , urate >9, urolithiasis, erosions, or tophi); goal urate <6; keep flare prophylaxis while titrating (ACR 2020).

Attribution & Changelog

Original: "Inflammatory Arthritis: A Pattern-Based Approach" — TeachIM.org, published December 2025.

Authors: Meredith Morcos, MD; Jennifer Stichman (Rheumatology, Univ. of Colorado). Executive editor: Brandon Fainstad, MD.

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Optimized 2026-06-11 (deck: Inflammatory-arthritis-differential.pptx; all 21 original interactive slides preserved).

Correction (slide 13, ankylosing spondylitis): imaging finding "bridging osteophytes" -> "syndesmophytes" (matches the lesson page; syndesmophytes are the characteristic axial-SpA lesion).

Currency check (June 2026): content verified against ACR 2020 gout guideline, CDC 2021 STI guidelines, IDSA 2020 Lyme guideline, IDSA/ASM 2024 microbiology guide — standing teaching unchanged. Note: for chlamydia co-treatment, CDC 2021 prefers doxycycline (azithromycin mainly in pregnancy).

Companion files: none published on the lesson page. Lesson package: HTML launcher, one-pager PDF, Anki cards.